Elementary Analysis

Chapter 1, Section 2

The Set of Rational Numbers
Exercises 2.1–2.8, with every subpart

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 1. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, letters in Section 3 denote real numbers. The empty sum is 0, and $0! = 1$.

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2 The set of rational numbers

The exercises in this section ask whether particular numbers belong to $\mathbb{Q}$. Recall that a number is rational when it can be written as m/n, where $m, n \in \mathbb{Z}$ and $n \neq 0$. A number is irrational when it is a real number that is not rational. The following elementary tools explain the steps used in the solutions.

Tool 1: Why rational arithmetic stays rational

Suppose $r = m/n$ and $s = p/q$, where $m, n, p, q \in \mathbb{Z}$ and $n, q \neq 0$. Putting fractions over a common denominator gives

$$r + s = \frac{mq + pn}{nq}, \qquad r - s = \frac{mq - pn}{nq}, \qquad rs = \frac{mp}{nq}.$$

Each numerator is an integer, and nq is a nonzero integer. Each displayed quantity is therefore rational by the definition of $\mathbb{Q}$. If $s \neq 0$, then $p \neq 0$, and

$$\frac{r}{s} = \frac{mq}{np}$$

also has an integer numerator and a nonzero integer denominator. Thus division by a nonzero rational number preserves rationality. Integers are rational because $j = j/1$. In particular, multiplication by an integer and subtraction of an integer preserve rationality.

For a positive integer k, repeated multiplication gives

$$r^k = \underbrace{r \cdot r \cdots r}_{k \text{ factors}} = \frac{m^k}{n^k} \in \mathbb{Q},$$

since $m^k, n^k \in \mathbb{Z}$ and $n^k \neq 0$. These formulas justify every use of "rational closure" below.

Tool 2: The integer cancellation fact behind the rational zeros theorem

Two integers are relatively prime when they have no common positive divisor other than 1. We use this elementary fact: if $u, v \in \mathbb{Z}$ are relatively prime, $v \neq 0$, and v divides uw, then v divides w. Here "v divides z" means that $z = vk$ for some $k \in \mathbb{Z}$.

For completeness, the cancellation fact follows from a short integer argument. The positive integers of the form $Au + Bv$, with $A, B \in \mathbb{Z}$, form a nonempty set: one of v and $-v$ is positive. Let d be the least such positive integer, and write $d = A_0u + B_0v$. The division algorithm gives $u = hd + t$, with $h \in \mathbb{Z}$ and $0 \leq t < d$. Then

$$t = u - hd = (1 - hA_0)u - hB_0v.$$

If $t > 0$, this would be a smaller positive integer of the same form, contradicting the choice of d. Hence $t = 0$, so d divides u. Applying the same argument to division of v by d shows that d divides v. Relative primality forces $d = 1$. Consequently there are integers A_0, B_0 with

$$A_0u + B_0v = 1.$$

Multiplying this identity by w gives

$$w = A_0(uw) + B_0vw.$$

If v divides uw, both terms on the right are integer multiples of v, so w is also an integer multiple of v. This proves the cancellation fact. The integer division algorithm and the existence of a least positive integer are the only integer facts used here.

One consequence needed below is worth spelling out. If u, v are relatively prime and v divides u^k, then cancellation in $u^k = u\,u^{k-1}$ shows that v divides u^{k-1}. Repeating this step eventually shows that v divides 1. When $u \neq 0$, the last power in this argument is $u^0 = 1$. When $u = 0$, relative primality already forces $|v| = 1$, so the same conclusion holds directly.

Tool 3: A monic polynomial can have only integer rational zeros

A polynomial is *monic* when the coefficient of its highest power is 1. Suppose

$$P(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0, \qquad a_0, \ldots, a_{d-1} \in \mathbb{Z},$$

where $d \geq 1$. Assume that $r \in \mathbb{Q}$ and $P(r) = 0$. Write $r = m/n$ in lowest terms, with $n > 0$. Thus $m, n \in \mathbb{Z}$ are relatively prime. If $m = 0$, then $r = 0$ is already an integer. For the denominator argument we may therefore assume $m \neq 0$. Substituting and multiplying by n^d gives

$$m^d + \sum_{j=0}^{d-1} a_j m^j n^{d-j} = 0.$$

Every term in the sum contains a factor n. Equivalently,

$$m^d = -n \left(\sum_{j=0}^{d-1} a_j m^j n^{d-j-1} \right).$$

The expression in parentheses is an integer, so n divides m^d. Tool 2, applied repeatedly to the relatively prime integers m, n, shows that n divides 1. As n is positive, $n = 1$. Therefore $r = m \in \mathbb{Z}$.

There is a further restriction if $a_0 \neq 0$. In that case $r \neq 0$, because $P(0) = a_0 \neq 0$. Substituting the integer m into $P(m) = 0$ and collecting all terms other than the constant gives

$$a_0 = -m \left(m^{d-1} + \sum_{j=1}^{d-1} a_j m^{j-1} \right).$$

For $d = 1$ the sum is empty and equals 0. In every case the expression in parentheses is an integer. Hence m divides a_0.

We have proved the monic case of Ross's Rational Zeros Theorem (Theorem 2.2 and Corollary 2.3):

> A rational zero of a monic polynomial with integer coefficients must be an integer. If the constant coefficient is nonzero, that integer must divide the constant coefficient.

In particular, if N is a positive integer and k is a positive integer, then a rational solution of $x^k = N$ must be an integer, because it is a zero of the monic integer polynomial $x^k - N$.

Tool 4: Bounding a positive root by consecutive integers

For $0 \leq u < v$ and a positive integer k,

$$v^k - u^k = (v - u)(v^{k-1} + v^{k-2}u + \cdots + u^{k-1}) > 0.$$

For $k = 1$ the factor in parentheses is 1. For $k \geq 2$, all its terms are nonnegative and its first term v^{k-1} is positive. Thus raising nonnegative numbers to the kth power preserves their strict order. This is an algebraic fact; no limiting argument is needed.

Let x be the nonnegative kth root of a positive integer N. Suppose a nonnegative integer a satisfies

$$a^k < N < (a+1)^k.$$

If $x \leq a$, then $x^k \leq a^k < N$, contradicting $x^k = N$. If $x \geq a+1$, then $x^k \geq (a+1)^k > N$, giving the same contradiction. It follows that

$$a < x < a+1.$$

There is no integer strictly between the consecutive integers a and $a+1$. Tool 3 therefore implies that x is irrational. This gives the criterion used in the first two exercises:

> If a positive integer lies strictly between two consecutive nonnegative integer kth powers, its positive kth root is irrational.

2.1 Exercise 2.1: Five square roots of integers

What we must show. Prove that $\sqrt{3}$, $\sqrt{5}$, $\sqrt{7}$, $\sqrt{24}$, and $\sqrt{31}$ are irrational.

Plan. Each number is a root of a monic integer polynomial $x^2 - N$. If it were rational, Tool 3 would make it an integer. We locate each root strictly between consecutive integers to rule that out.

1. Record the exact integer-square comparisons

The needed computations are

$$1^2 = 1, \quad 2^2 = 4, \quad 3^2 = 9, \quad 4^2 = 16, \quad 5^2 = 25, \quad 6^2 = 36.$$

They give the following bounds by Tool 4:

N	Consecutive square bounds	Consequent root bounds
3	$1^2 = 1 < 3 < 4 = 2^2$	$1 < \sqrt{3} < 2$
5	$2^2 = 4 < 5 < 9 = 3^2$	$2 < \sqrt{5} < 3$
7	$2^2 = 4 < 7 < 9 = 3^2$	$2 < \sqrt{7} < 3$
24	$4^2 = 16 < 24 < 25 = 5^2$	$4 < \sqrt{24} < 5$
31	$5^2 = 25 < 31 < 36 = 6^2$	$5 < \sqrt{31} < 6$

2. Apply the rational-root argument to each number

(a) Suppose $\sqrt{3}$ were rational. Since $(\sqrt{3})^2 - 3 = 0$, Tool 3 applied to $x^2 - 3$ would imply $\sqrt{3} \in \mathbb{Z}$. But $1 < \sqrt{3} < 2$, and no integer satisfies those two inequalities. This contradiction proves $\sqrt{3} \notin \mathbb{Q}$.

(b) Suppose $\sqrt{5}$ were rational. The equation $(\sqrt{5})^2 - 5 = 0$ and Tool 3 imply $\sqrt{5} \in \mathbb{Z}$. The bounds $2 < \sqrt{5} < 3$ exclude every integer, so $\sqrt{5} \notin \mathbb{Q}$.

(c) Suppose $\sqrt{7}$ were rational. The monic integer equation $(\sqrt{7})^2 - 7 = 0$ would make $\sqrt{7}$ an integer. The strict bounds $2 < \sqrt{7} < 3$ make this impossible. Hence $\sqrt{7} \notin \mathbb{Q}$.

(d) Suppose $\sqrt{24}$ were rational. Because it satisfies $x^2 - 24 = 0$, it would be an integer by Tool 3. Its bounds $4 < \sqrt{24} < 5$ contradict that conclusion. Thus $\sqrt{24} \notin \mathbb{Q}$.

(e) Suppose $\sqrt{31}$ were rational. Tool 3 applied to x^2-31 would make it an integer. Since $5 < \sqrt{31} < 6$, there is no such integer. Therefore $\sqrt{31} \notin \mathbb{Q}$.

3. Establish one auxiliary case for later exercises

We will also need that $\sqrt{2}$ is irrational. If it were rational, the equation $(\sqrt{2})^2 - 2 = 0$ would make it an integer. But

$$1^2 = 1 < 2 < 4 = 2^2 \quad \Longrightarrow \quad 1 < \sqrt{2} < 2,$$

which is impossible for an integer. Thus $\sqrt{2} \notin \mathbb{Q}$ as well.

Conclusion. All five requested square roots are irrational. The same argument also proves the auxiliary fact $\sqrt{2} \notin \mathbb{Q}$.

2.2 Exercise 2.2: Three higher roots of integers

What we must show. Prove that $\sqrt[3]{2}$, $\sqrt[7]{5}$, and $\sqrt[4]{13}$ are irrational.

Plan. Tool 3 applies to $x^3 - 2$, $x^7 - 5$, and $x^4 - 13$. We compute suitable consecutive integer powers, then apply Tool 4.

1. The cube root of 2

Let $x = \sqrt[3]{2}$. By the definition of the positive cube root, $x > 0$ and $x^3 = 2$. Assume that $x \in \mathbb{Q}$. Since x is a zero of the monic polynomial $x^3 - 2$, whose coefficients are integers, Tool 3 gives $x \in \mathbb{Z}$.

Now $1^3 = 1$ and $2^3 = 2 \cdot 2 \cdot 2 = 8$. Consequently

$$1^3 = 1 < 2 < 8 = 2^3.$$

Tool 4 gives $1 < x < 2$. No integer lies in that interval, contradicting $x \in \mathbb{Z}$. Hence $\sqrt[3]{2}$ is irrational.

2. The seventh root of 5

Let $y = \sqrt[7]{5}$. Then $y > 0$ and $y^7 = 5$. If y were rational, Tool 3 applied to $x^7 - 5$ would imply $y \in \mathbb{Z}$. The exact powers are

$$1^7 = 1, \qquad 2^7 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 128.$$

Thus

$$1^7 = 1 < 5 < 128 = 2^7,$$

so $1 < y < 2$. This excludes all integers and contradicts the consequence of the rationality assumption. Therefore $\sqrt[7]{5}$ is irrational.

3. The fourth root of 13

Let $z = \sqrt[4]{13}$, where the fourth-root notation denotes the positive real root. Then $z > 0$ and $z^4 = 13$. If $z \in \mathbb{Q}$, the monic integer equation $z^4 - 13 = 0$ gives $z \in \mathbb{Z}$ by Tool 3. However,

$$1^4 = 1 < 13 < 16 = 2 \cdot 2 \cdot 2 \cdot 2 = 2^4.$$

It follows that $1 < z < 2$, contradicting integer membership. Hence $\sqrt[4]{13}$ is irrational.

Conclusion. The numbers $\sqrt[3]{2}$, $\sqrt[7]{5}$, and $\sqrt[4]{13}$ are all irrational.

2.3 Exercise 2.3: A nested square root

What we must show. Prove that $\sqrt{2+\sqrt{2}}$ is irrational.

Plan. Assume the outer square root is rational. Square it and isolate $\sqrt{2}$. Rational closure would then force $\sqrt{2}$ to be rational, contradicting the auxiliary case in Exercise 2.1.

1. Name the number and record its defining equation

Put

$$a = \sqrt{2+\sqrt{2}}.$$

The inner root $\sqrt{2}$ is positive, so $2+\sqrt{2} > 0$. The outer square root therefore denotes a positive real number. Its defining equation is

$$a^2 = 2+\sqrt{2}.$$

2. Assume rationality and rearrange

Suppose, for a contradiction, that $a \in \mathbb{Q}$. Then $a^2 \in \mathbb{Q}$ by rational closure under multiplication. Subtracting the rational integer 2 from both sides gives

$$\sqrt{2} = a^2 - 2 \in \mathbb{Q}.$$

In fraction form, if $a = m/n$ with $m, n \in \mathbb{Z}$ and $n \neq 0$, this is

$$a^2 - 2 = \frac{m^2}{n^2} - \frac{2n^2}{n^2} = \frac{m^2-2n^2}{n^2},$$

an integer divided by a nonzero integer. Thus the rationality conclusion is justified directly by the definition, as well as by Tool 1.

3. Identify the contradiction

Exercise 2.1 established $\sqrt{2} \notin \mathbb{Q}$. The rationality assumption for a has produced its negation, $\sqrt{2} \in \mathbb{Q}$. Therefore the assumption is false.

Conclusion. $\sqrt{2+\sqrt{2}}$ is irrational.

2.4 Exercise 2.4: A nested cube root

What we must show. Prove that $\sqrt[3]{5-\sqrt{3}}$ is irrational.

Plan. Cube the number to remove the outer radical, then isolate $\sqrt{3}$. If the original number were rational, this rearrangement would express $\sqrt{3}$ using rational operations only.

1. Check the sign and write the cube equation

Exercise 2.1 gives $1 < \sqrt{3} < 2$. In particular,

$$5 - \sqrt{3} > 5 - 2 = 3 > 0.$$

Thus

$$a = \sqrt[3]{5-\sqrt{3}}$$

is a positive real number, and the definition of cube root gives

$$a^3 = 5 - \sqrt{3}.$$

2. Derive a rational expression for $\sqrt{3}$ under the assumption

Assume $a \in \mathbb{Q}$. Then $a^3 \in \mathbb{Q}$ by Tool 1. Adding $\sqrt{3}$ to the cube equation and subtracting a^3 gives

$$a^3 + \sqrt{3} = 5, \qquad \text{hence} \qquad \sqrt{3} = 5 - a^3.$$

Both 5 and a^3 are rational, so their difference is rational. Explicitly, if $a = m/n$ with integer m, n and $n \neq 0$, then

$$5 - a^3 = \frac{5n^3 - m^3}{n^3} \in \mathbb{Q}.$$

3. Finish the contradiction

The preceding step gives $\sqrt{3} \in \mathbb{Q}$, contrary to Exercise 2.1. Therefore $a \notin \mathbb{Q}$.

Conclusion. $\sqrt[3]{5 - \sqrt{3}}$ is irrational.

2.5 Exercise 2.5: A positive real fractional power

What we must show. Prove that $(3 + \sqrt{2})^{2/3}$ is irrational.

Plan. First explain the fractional exponent using the positive real cube root. Cubing the proposed number then produces a simple expression $11 + 6\sqrt{2}$, from which $\sqrt{2}$ can be isolated.

1. Interpret the exponent without a sign ambiguity

Since $\sqrt{2} > 0$, the base

$$u = 3 + \sqrt{2}$$

is positive. Let $t = \sqrt[3]{u} > 0$. Thus $t^3 = u$, and the positive real fractional power in the problem is

$$a = u^{2/3} = (\sqrt[3]{u})^2 = t^2.$$

It follows from integer power rules that

$$a^3 = (t^2)^3 = t^6 = (t^3)^2 = u^2 = (3 + \sqrt{2})^2.$$

This explains precisely why cubing removes the exponent $2/3$.

2. Expand the square completely

Using $(v + w)^2 = v^2 + 2vw + w^2$ gives

$$\begin{aligned} a^3 &= (3 + \sqrt{2})^2 \\ &= 3^2 + 2 \cdot 3 \cdot \sqrt{2} + (\sqrt{2})^2 \\ &= 9 + 6\sqrt{2} + 2 \\ &= 11 + 6\sqrt{2}. \end{aligned}$$

Subtracting 11 and dividing by 6 therefore yields

$$a^3 - 11 = 6\sqrt{2}, \qquad \sqrt{2} = \frac{a^3 - 11}{6}.$$

3. Use rational closure to obtain the contradiction

Suppose $a \in \mathbb{Q}$. Then $a^3 \in \mathbb{Q}$, hence $a^3 - 11 \in \mathbb{Q}$. Because 6 is a nonzero rational integer, division by 6 preserves rationality. The last displayed equation would imply $\sqrt{2} \in \mathbb{Q}$. This contradicts the auxiliary irrationality result in Exercise 2.1.

Conclusion. $(3 + \sqrt{2})^{2/3}$, with the positive real interpretation of the fractional power, is irrational.

2.6 Exercise 2.6: A rational quadratic expression

What we must show. Explain why $b \in \mathbb{Q}$ implies $4 - 7b^2 \in \mathbb{Q}$.

Plan. Start with the defining fraction for a rational number and write $4 - 7b^2$ as one fraction. Check its numerator and denominator separately.

1. Write the rationality hypothesis explicitly

Assume $b \in \mathbb{Q}$. By definition, there are integers m, n such that

$$b = \frac{m}{n}, \qquad n \neq 0.$$

The numerator m may be positive, negative, or zero. No restriction on the sign of b is needed for this argument.

2. Perform the arithmetic over one denominator

Squaring and multiplying gives

$$b^2 = \frac{m^2}{n^2}, \qquad 7b^2 = \frac{7m^2}{n^2}.$$

As $n \neq 0$, its square n^2 is nonzero, so we may also write $4 = 4n^2/n^2$. Subtracting now gives

$$4 - 7b^2 = \frac{4n^2}{n^2} - \frac{7m^2}{n^2} = \frac{4n^2 - 7m^2}{n^2}.$$

The integer operations of squaring, multiplication, and subtraction show that $4n^2 - 7m^2 \in \mathbb{Z}$. The denominator n^2 is a nonzero integer. The displayed fraction is therefore rational by definition.

3. Explain the connection with Example 6

The same reasoning can be summarized using Tool 1: $b \in \mathbb{Q}$ implies $b^2 \in \mathbb{Q}$, then $7b^2 \in \mathbb{Q}$, and then $4 - 7b^2 \in \mathbb{Q}$. This is the rational-closure step used in Example 6. For the particular number considered there, the radical equation also gives

$$7b^2 = 4 - 2\sqrt{3}, \qquad \text{so} \qquad 4 - 7b^2 = 2\sqrt{3}.$$

If that b were rational, the result just proved would make $2\sqrt{3}$ rational. Dividing by the nonzero rational number 2 would make $\sqrt{3}$ rational, contradicting Exercise 2.1. This shows how the closure statement supplies the needed step in an irrationality proof.

Conclusion. For every rational $b = m/n$ with $n \neq 0$, $4 - 7b^2 = (4n^2 - 7m^2)/n^2 \in \mathbb{Q}$.

2.7 Exercise 2.7: Two radicals that simplify to rational numbers

What we must show. Show that

$$\text{(a)} \quad \sqrt{4+2\sqrt{3}}-\sqrt{3}, \qquad \text{(b)} \quad \sqrt{6+4\sqrt{2}}-\sqrt{2}$$

are rational, and find their values. In each expression the final subtracted square root lies outside the first radical.

Plan. Recognize each radicand as the square of a positive sum. Positivity is essential: for a real v, the principal square root satisfies $\sqrt{v^2}=|v|$. To replace that absolute value by v, we check that $v>0$.

1. Part (a): Expand the proposed positive square root

Consider $v=1+\sqrt{3}$. Because $\sqrt{3}>0$, we have $v>0$. Expanding its square gives

$$\begin{aligned}(1+\sqrt{3})^2 &= 1^2+2\cdot 1\cdot\sqrt{3}+(\sqrt{3})^2\\ &= 1+2\sqrt{3}+3\\ &= 4+2\sqrt{3}.\end{aligned}$$

The radicand is positive, and $1+\sqrt{3}$ is a positive number whose square equals that radicand. It is therefore the principal square root:

$$\sqrt{4+2\sqrt{3}}=1+\sqrt{3}.$$

One can also justify the choice of sign from uniqueness. If two nonnegative numbers v, w have the same positive square, then $(v-w)(v+w)=v^2-w^2=0$. Since $v+w>0$, it follows that $v-w=0$, so $v=w$. There is thus only one nonnegative square root here.

Substituting the identity into the expression in part (a) gives

$$\sqrt{4+2\sqrt{3}}-\sqrt{3}=(1+\sqrt{3})-\sqrt{3}=1+(\sqrt{3}-\sqrt{3})=1+0=1.$$

As $1=1/1$, this value is rational.

2. Part (b): Expand the second proposed positive square root

Now consider $w=2+\sqrt{2}$. It is positive because $\sqrt{2}>0$. Its square expands as

$$\begin{aligned}(2+\sqrt{2})^2 &= 2^2+2\cdot 2\cdot\sqrt{2}+(\sqrt{2})^2\\ &= 4+4\sqrt{2}+2\\ &= 6+4\sqrt{2}.\end{aligned}$$

Thus $2+\sqrt{2}$ is the positive square root of $6+4\sqrt{2}$. The sign check permits the identity

$$\sqrt{6+4\sqrt{2}}=2+\sqrt{2}.$$

It follows that

$$\sqrt{6+4\sqrt{2}}-\sqrt{2}=(2+\sqrt{2})-\sqrt{2}=2+(\sqrt{2}-\sqrt{2})=2+0=2.$$

Since $2=2/1$, this value is rational as well.

Conclusion. Part (a) equals 1 and part (b) equals 2; both expressions are rational numbers.

2.8 Exercise 2.8: All rational zeros of an eighth-degree polynomial

What we must show. Find every rational solution of

$$x^8 - 4x^5 + 13x^3 - 7x + 1 = 0.$$

Plan. The polynomial is monic, has integer coefficients, and has constant term 1. The monic rational-zero result restricts every rational root to an integer divisor of 1. We must test both candidates and explain why this exhausts the rational possibilities.

1. Obtain the complete candidate list

Let

$$P(x) = x^8 - 4x^5 + 13x^3 - 7x + 1.$$

The coefficient of x^8 is 1, so P is monic. Its missing powers have coefficient 0, which is also an integer. Its constant coefficient is $1 \neq 0$. Thus Tool 3 applies to every rational zero of P. It says that such a zero must be an integer dividing 1. If an integer m divides 1, then $1 = mk$ for some integer k. Taking absolute values gives $1 = |m|\,|k|$, so both nonnegative integer factors must equal 1. In particular $|m| = 1$, and hence $m = 1$ or $m = -1$. The full rational candidate list is therefore

$$\{1, -1\}.$$

Here is the denominator step directly for this particular polynomial. If a rational root is m/n in lowest terms, with $n > 0$, multiplication of $P(m/n) = 0$ by n^8 gives

$$m^8 - 4m^5n^3 + 13m^3n^5 - 7mn^7 + n^8 = 0.$$

Solving for m^8 gives

$$m^8 = n(4m^5n^2 - 13m^3n^4 + 7mn^6 - n^7),$$

so n divides m^8. Since m, n are relatively prime, repeated coprime cancellation gives $n \mid 1$, hence $n = 1$. After this reduction, $P(m) = 0$ can be rearranged as

$$1 = -m^8 + 4m^5 - 13m^3 + 7m = m(-m^7 + 4m^4 - 13m^2 + 7).$$

This proves $m \mid 1$ and recovers precisely the two candidates above. Also $P(0) = 1$, so zero has not been omitted as a possible root.

2. Evaluate the candidate 1

Every positive power of 1 equals 1. Consequently

$$\begin{aligned} P(1) &= 1^8 - 4 \cdot 1^5 + 13 \cdot 1^3 - 7 \cdot 1 + 1 \\ &= 1 - 4 + 13 - 7 + 1 \\ &= -3 + 13 - 7 + 1 \\ &= 10 - 7 + 1 \\ &= 3 + 1 = 4. \end{aligned}$$

Since $4 \neq 0$, the number 1 is not a solution.

3. Evaluate the candidate -1

The eighth power of -1 is 1, while its fifth and third powers are -1. Substituting with the signs retained gives

$$\begin{aligned} P(-1) &= (-1)^8 - 4(-1)^5 + 13(-1)^3 - 7(-1) + 1 \\ &= 1 - 4(-1) + 13(-1) + 7 + 1 \\ &= 1 + 4 - 13 + 7 + 1 \\ &= 5 - 13 + 7 + 1 \\ &= -8 + 7 + 1 \\ &= -1 + 1 = 0. \end{aligned}$$

Thus -1 is a solution, and it is rational because $-1 = (-1)/1$.

4. Explain why the search is complete

Every rational solution must belong to the candidate set $\{1, -1\}$ from Step 1. Step 2 rejects 1, and Step 3 verifies -1. There are no other rational candidates to examine.

Conclusion. The set of all rational solutions is $\{-1\}$.